A Quantum Register Model for Correlated Market Reactions

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Abstract

Recent work explores quantum financial modeling, but many examples do not clearly explain how events impact the market. In this project, there is a nine-qubit quantum circuit to map simplified event features to directional probabilities for major assets such as USD, Gold, the S&P 500, and Bitcoin. The model encodes event intensity and category scores, then uses 34 trainable parameters to tune event-asset and asset-asset relationships. Predictions were evaluated using real world price data and qiskit simulation. This project presents an interpretable educational demonstration of quantum finance, not a real trading or forecasting system.

Background

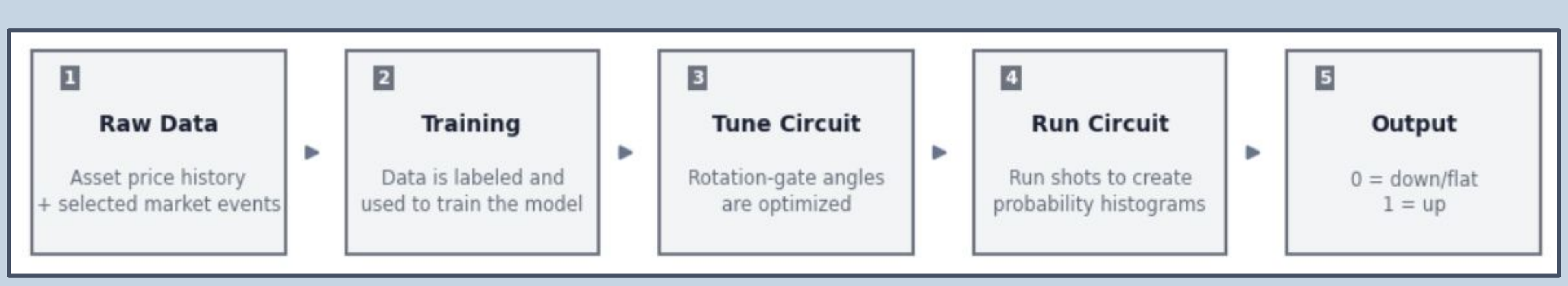
Quantum computers use qubits, which can exist in a superposition of 0 and 1 before measurement. Quantum circuits use rotation and entangling gates to reshape the probability of different measurement outcomes.

In a parameterized quantum circuit, rotation gate angles are treated as tunable parameters. During training, these parameters are adjusted so the circuit's output probabilities better match patterns in the data.

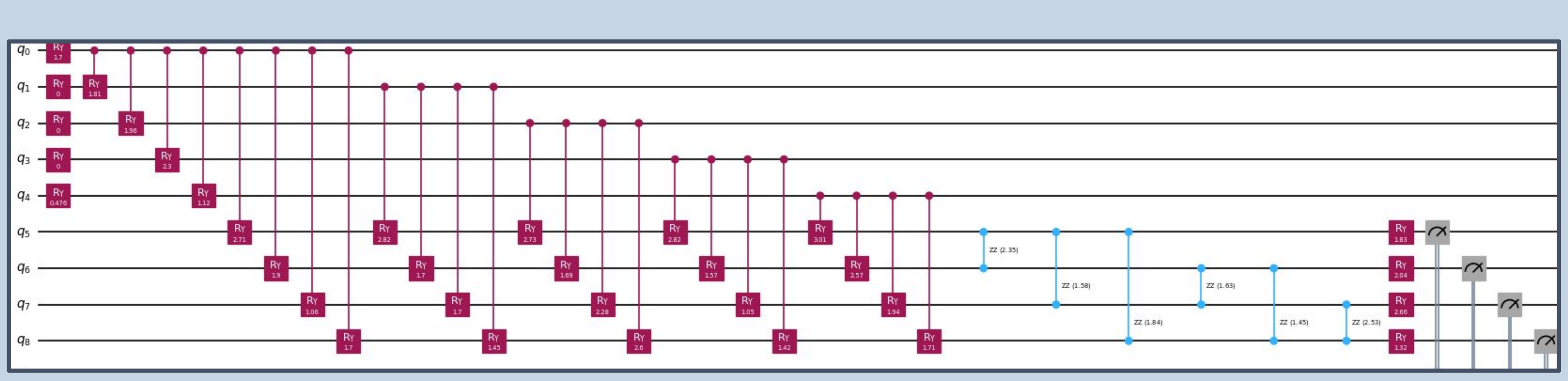
In this project, event scores are encoded into input qubits, then connected to asset qubits representing USD, Gold, S&P 500, and Bitcoin. The circuit outputs up/down probabilities for each asset as an educational market-modeling example, not a tradable forecasting system.

Model/Methods

Asset price history and selected market events are combined into a labeled training dataset. Each event is converted into five scores representing intensity, macro/geopolitical pressure, disaster disruption, technology adoption, and regulation/policy impact.

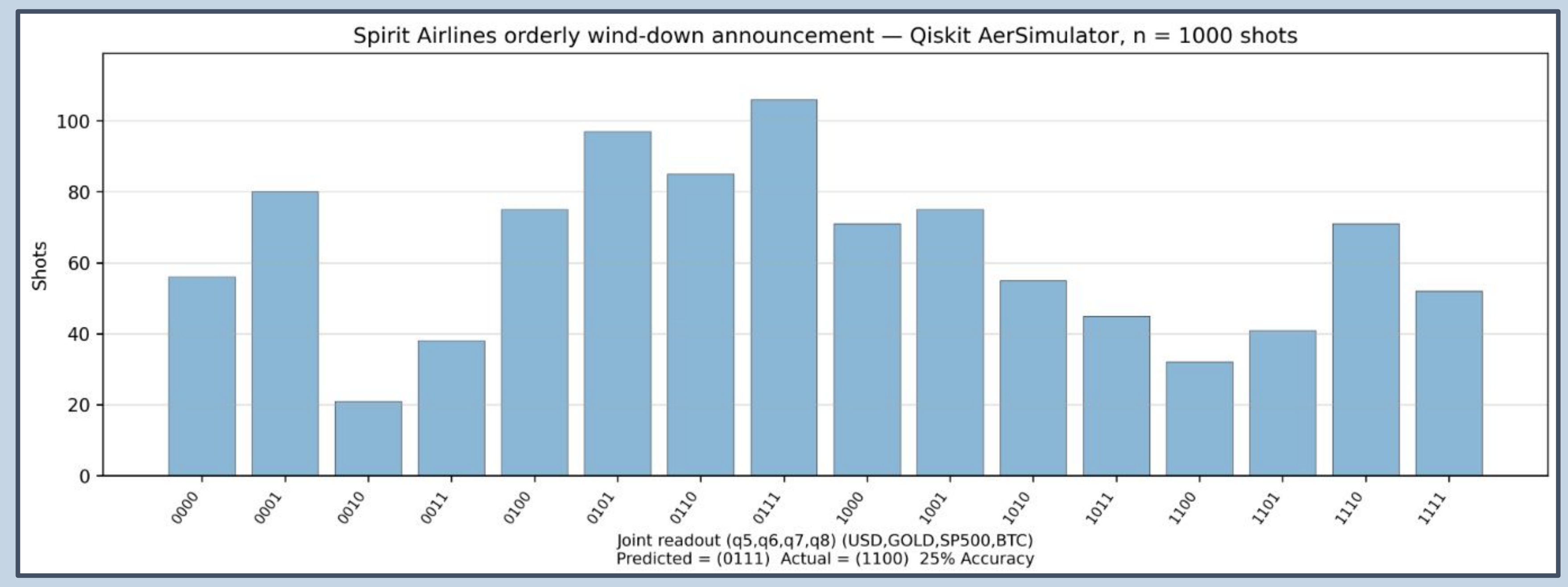


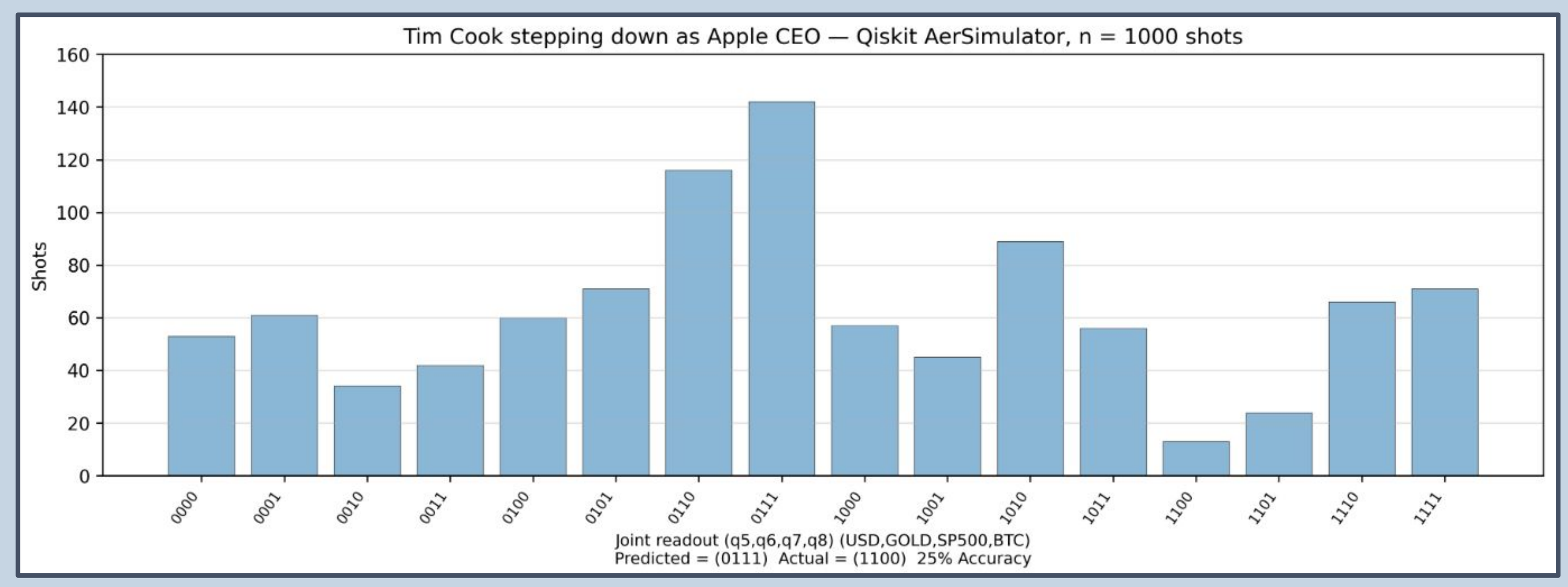
These five event scores are encoded into input qubits q0–q4 of a 9-qubit quantum circuit. Four additional qubits, q5–q8, represent the asset outputs: USD, Gold, S&P 500, and Bitcoin.

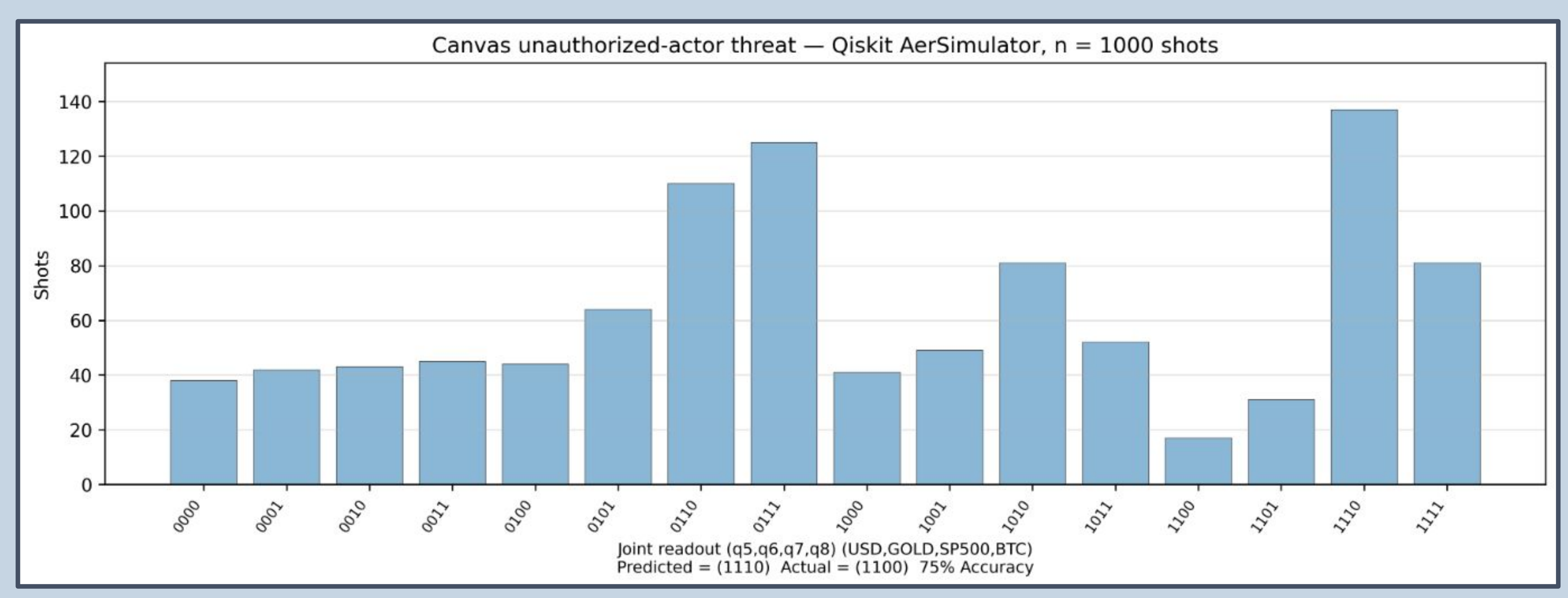


Results/Discussion

The trained circuit was tested on three recent events that were not included in training. For each event, the model produced probability histograms for whether each asset would move up or down over the next five trading days. The 3 events included:







The model correctly predicted 5 out of 12 asset directions, or about 41.7% accuracy across the three events. This shows that the circuit was able to run on unseen events and produce interpretable probability outputs, but its predictive accuracy was limited.

The mixed results are expected because the dataset is small, the event scores are simplified, and market movements depend on many factors not included in the model. Overall, the project demonstrates a working quantum machine-learning pipeline, but not a reliable trading forecast.

Conclusions/Future Work

This model is not reliable for predicting prices on its own, but future versions could improve by adding registers for price-movement trends and using better event data, labeling, and larger training sets.

References

[1] Qiskit Community, "Qiskit: An open-source framework for quantum computing," Zenodo, 2017. doi: 10.5281/zenodo.2562110.